MEASUREMENTS AND INSTRUMENTS - COMPLETE NOTES

1. FUNDAMENTALS OF MEASUREMENT

1.1 Definition

Measurement is the process of comparing an unknown quantity with a known fixed standard quantity.

- **Measurand:** The physical quantity being measured (e.g., pressure, temperature, voltage)
- **Unit:** The standard reference (e.g., meter, second, Volt)
- **Instrument:** The device used to perform the measurement

1.2 Types of Measurement

Type	Description	Example
Direct Measurement	Value obtained directly from instrument	Measuring length with ruler
Indirect Measurement	Value derived from other measurements	Resistance = V/I using ammeter and voltmeter
Null Measurement	Balanced condition indicates value	Wheatstone bridge
Deflection Measurement	Pointer movement indicates value	Analog voltmeter
Absolute Measurement	Based on physical constants	Tangent galvanometer

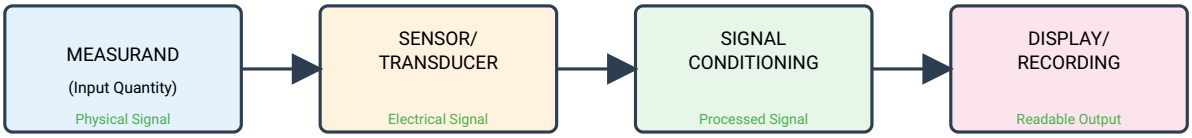
1.3 Fundamental vs Derived Units

Fundamental Units (SI Base Units)		
Quantity	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric Current	ampere	A
Temperature	kelvin	K
Luminous Intensity	candela	cd
Amount of Substance	mole	mol

Derived Units			
Quantity	Unit	Symbol	Expression

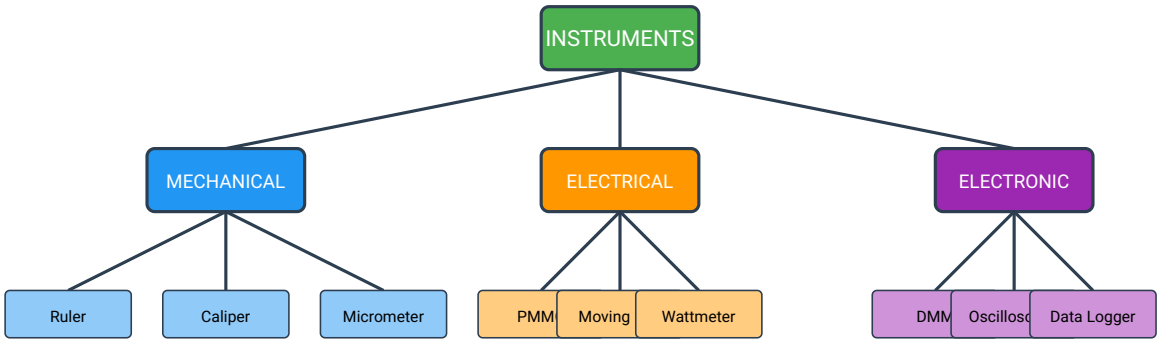
Force	newton	N	kg·m/s²
Pressure	pascal	Pa	N/m²
Energy	joule	J	N·m
Power	watt	W	J/s
Voltage	volt	V	W/A
Resistance	ohm	Ω	V/A

Figure 1: Generalized Measurement System



Block diagram showing the three main stages of a measurement system

Figure 2: Instrument Classification Tree

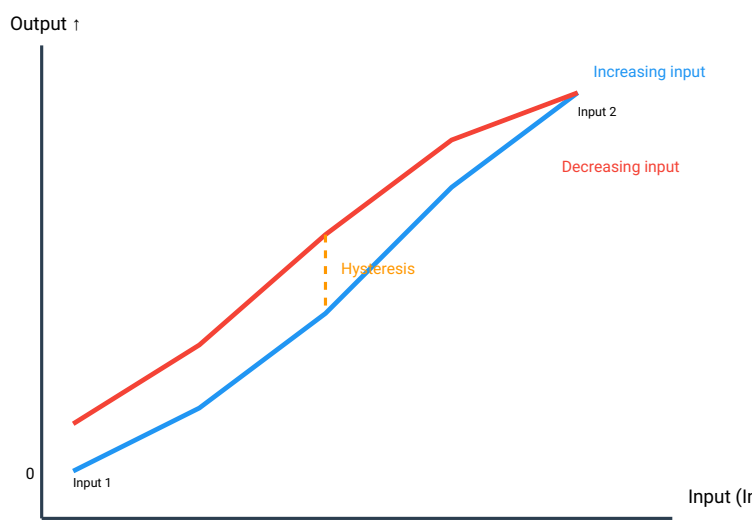


2. INSTRUMENT CHARACTERISTICS

2.1 Static Characteristics

Term	Definition	Formula
Accuracy	Closeness to true value	$A = 1 - \lvert(V_m - V_t)/V_t\rvert \times 100\%$
Precision	Closeness of repeated measurements	$\sigma = \sqrt{[\Sigma(x_i - \mu)^2/n]}$
Sensitivity	Output change per input change	$S = \Delta Output/\Delta Input$
Resolution	Smallest detectable change	Smallest scale division
Hysteresis	Output difference for increasing vs decreasing input	$H = \lvert O_{up} - O_{down}\rvert$
Linearity	Deviation from ideal straight line	$\%NL = (\Delta max/FS) \times 100\%$
Drift	Gradual change over time	μV/°C or μV/hour

Figure 3: Hysteresis Curve



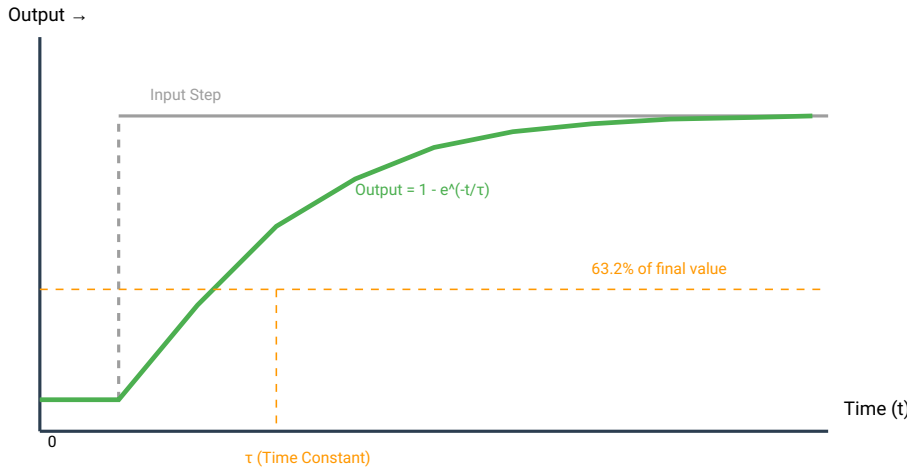
Hysteresis shows the difference in output when input is increasing versus decreasing

2.2 Dynamic Characteristics

Term	Definition	Formula (First Order System)
Time Constant (τ)	Time to reach 63.2% of final value	$\tau = RC$ or $\tau = L/R$
Rise Time (tr)	Time from 10% to 90% of final value	$t_r = 2.2\tau$

Settling Time (ts)	Time to stay within tolerance band	ts = 4τ (2%), ts = 5τ (5%)
Overshoot (Mp)	Maximum peak exceeding steady-state	Mp = e^(-πζ/√(1-ζ²)) × 100%

Figure 4: First Order System Response (Step Input)



3. ERRORS IN MEASUREMENT

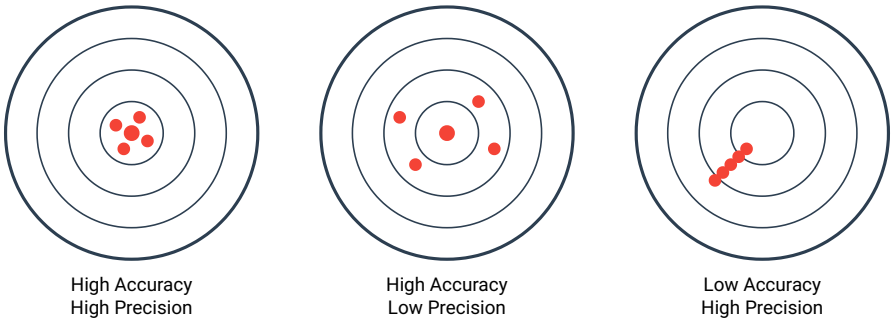
3.1 Types of Errors

Error Type	Description	Examples	Correction
Gross Errors	Human mistakes, blunders	Misreading scale, incorrect recording, improper use	Careful operation, multiple readings
Systematic Errors	Consistent, repeatable errors	Calibration error, zero error, loading effect	Calibration, correction factors
Random Errors	Unpredictable variations	Noise, friction, temperature fluctuations	Statistical analysis, averaging
Environmental Errors	Due to external conditions	Temperature, humidity, pressure, magnetic fields	Environmental control, compensation
Instrumental Errors	Due to instrument limitations	Hysteresis, nonlinearity, drift	Calibration, better instruments
Observational Errors	Due to observer	Parallax error, reaction time	Proper technique, digital readouts

```
Error Formulas:
Absolute Error = |Measured Value - True Value|
```

```
Relative Error = (Absolute Error / True Value) × 100%
Correction Factor = True Value - Measured Value
```

Figure 5: Accuracy vs Precision Illustration



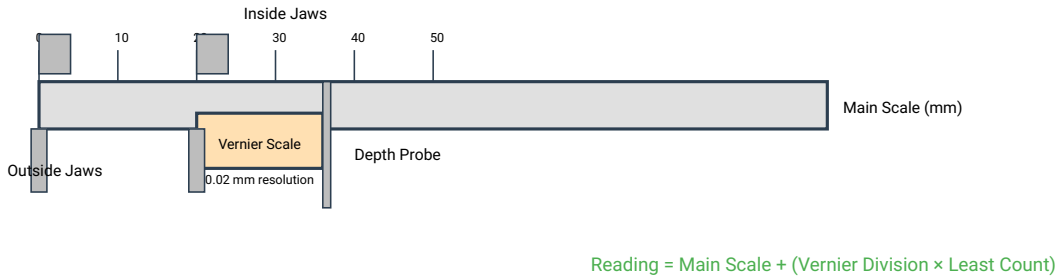
Center = True Value | Dots = Measurements

4. MECHANICAL INSTRUMENTS

4.1 Linear Measurement Instruments

Instrument	Range	Resolution	Principle	Application
Steel Rule	150-1000 mm	0.5 mm / 1 mm	Direct reading	General measurement
Vernier Caliper	0-150 mm / 0-300 mm	0.02 mm / 0.05 mm	Vernier scale principle	ID, OD, depth measurement
Micrometer Screw Gauge	0-25 mm / 25-50 mm	0.01 mm	Screw thread principle	Precision measurement
Dial Gauge	0-10 mm / 0-50 mm	0.01 mm	Rack and pinion	Runout, alignment
Height Gauge	0-300 mm / 0-600 mm	0.02 mm	Vernier scale	Layout, inspection
Slip Gauges	Set of blocks	0.001 mm	Wringing	Calibration, precision setup

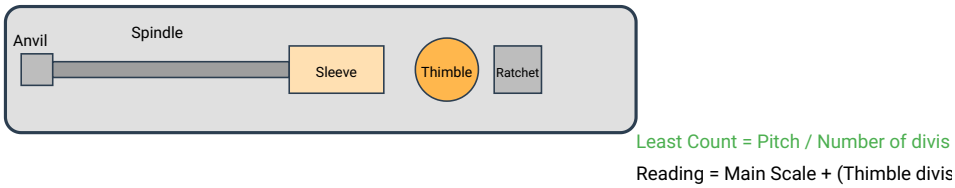
Figure 6: Vernier Caliper



4.2 Angular Measurement Instruments

Instrument	Range	Resolution	Principle
Bevel Protractor	0-360°	5 minutes (5')	Vernier principle on circular scale
Sine Bar	0-90°	Seconds	Trigonometric principle: sin θ = h/L
Angle Gauge	Set of angles	Minutes	Wringing of angle blocks
Clinometer	0-180°	1 minute	Spirit level with protractor

Figure 7: Micrometer Screw Gauge



5. ELECTRICAL INSTRUMENTS

5.1 Analog Instruments

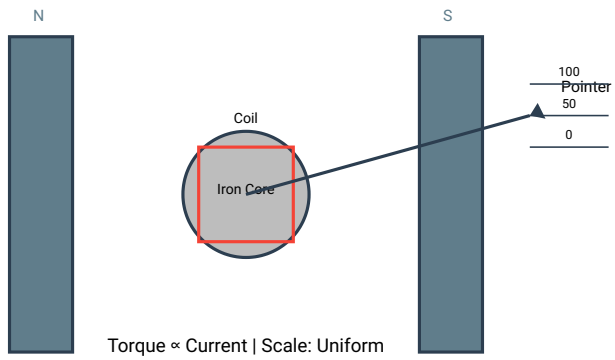
PMMC (Permanent Magnet Moving Coil) Instrument

- **Principle:** When current passes through a coil placed in magnetic field, a torque is produced proportional to current
- **Advantages:** High accuracy, uniform scale, low power consumption
- **Disadvantages:** Only for DC measurement, expensive
- **Application:** DC voltmeter, DC ammeter, galvanometer

Moving Iron (MI) Instrument

- **Principle:** A piece of iron moves when placed in magnetic field of a coil carrying current
- **Advantages:** Can measure AC and DC, robust, cheap
- **Disadvantages:** Non-uniform scale, hysteresis error
- **Application:** AC voltmeter, AC ammeter, panel meters

Figure 8: PMMC Instrument Construction

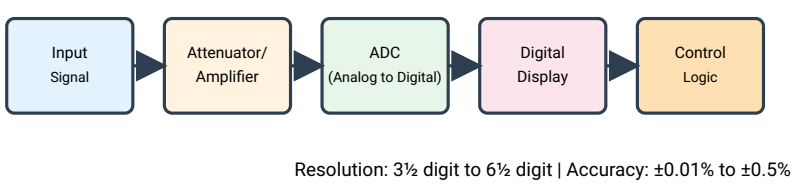


5.2 Digital Instruments

Digital Multimeter (DMM)

- **Principle:** Converts analog input to digital using ADC, displays numeric value
- **Features:** Measures voltage, current, resistance, continuity, capacitance, frequency
- **Advantages:** High accuracy, no parallax error, high input impedance, auto-ranging

Figure 9: Digital Multimeter Block Diagram

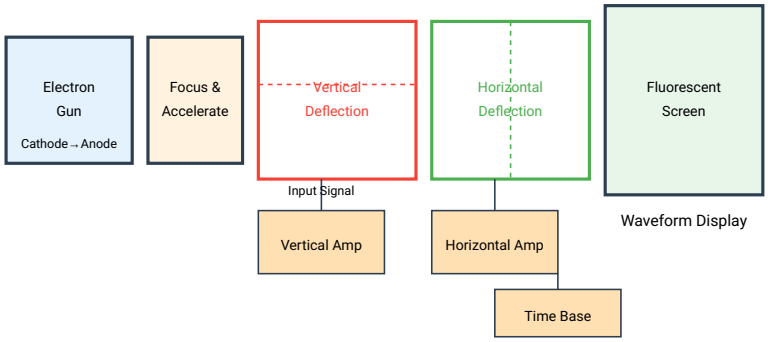


6. ELECTRONIC INSTRUMENTS

6.1 Cathode Ray Oscilloscope (CRO)

Principle: Electron beam is deflected by electric fields to display voltage vs time waveform on fluorescent screen.

Figure 10: Cathode Ray Oscilloscope Block Diagram



6.2 Digital Storage Oscilloscope (DSO)

- Captures and stores waveforms digitally
- Allows analysis, zooming, and measurement of stored signals
- Provides FFT analysis, automated measurements, and data export

7. TRANSDUCERS & SENSORS

Transducer: A device that converts one form of energy to another. A sensor converts physical quantity to electrical signal.

7.1 Classification of Transducers

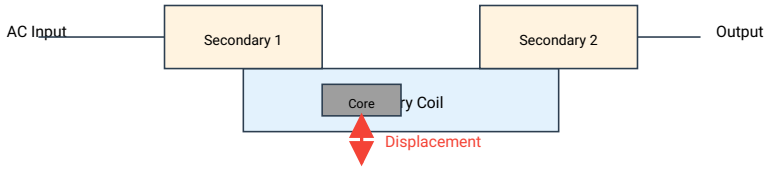
- **Active vs Passive:** Active generates output (thermocouple), Passive requires external excitation (strain gauge)
- **Analog vs Digital:** Analog gives continuous output, Digital gives discrete output
- **Primary vs Secondary:** Primary directly responds to measurand, Secondary converts primary output

7.2 Common Transducers

Measurand	Transducer Type	Working Principle	Output
Displacement	LVDT	Mutual inductance change with core position	AC Voltage
Displacement	Potentiometer	Resistance change with wiper position	Resistance
Strain	Strain Gauge	Piezoresistive effect	Resistance change
Temperature	Thermocouple	Seebeck effect	Voltage (mV)
Temperature	RTD (Pt100)	Resistance increases with temperature	Resistance

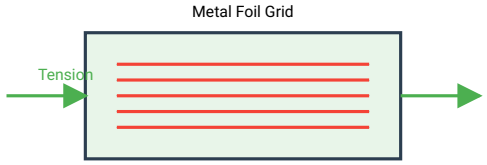
Temperature	Thermistor	Resistance decreases with temperature (NTC)	Resistance
Pressure	Bourdon Tube	Elastic deformation	Mechanical motion
Pressure	Diaphragm + Strain Gauge	Strain from pressure deflection	Resistance
Pressure	Piezoelectric	Piezoelectric effect	Charge/Voltage
Flow	Orifice Plate	Pressure drop across restriction	Pressure difference
Flow	Electromagnetic Flowmeter	Faraday's law of induction	Voltage
Light	Photodiode	Light generates current	Current
Light	LDR (Photoresistor)	Resistance decreases with light	Resistance
Acceleration	Accelerometer	Piezoelectric/MEMS	Charge/Voltage
Humidity	Hygrometer	Capacitance change with moisture	Capacitance

Figure 11: LVDT (Linear Variable Differential Transformer)



Output Voltage ∝ Core Displacement | Phase indicates direction

Figure 12: Strain Gauge Principle



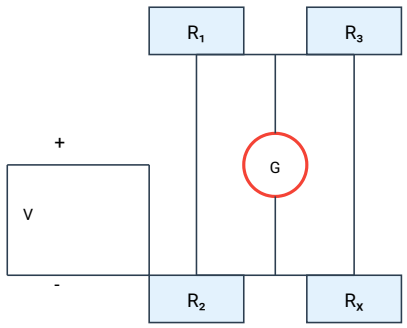
$\Delta R/R = G \times \varepsilon$ | where G = Gauge Factor, ε = Strain

8. BRIDGES & MEASURING CIRCUITS

8.1 Wheatstone Bridge

Principle: Null-type bridge used for precise measurement of resistance.

Figure 13: Wheatstone Bridge Circuit



At Balance: $R_1/R_2 = R_3/R_x$

$R_x = (R_2 \times R_3)/R_1$

8.2 Other Bridges

Bridge	Measured Quantity	Balance Condition	Application
Wheatstone Bridge	Resistance (DC)	$R_1/R_2 = R_3/R_x$	Precision resistance measurement
Kelvin Bridge	Low Resistance	Modified Wheatstone	Low resistance (μΩ to Ω)
Maxwell Bridge	Inductance	$R_1/R_2 = R_3/R_x$	Inductance measurement
Hay Bridge	Inductance (High Q)	Complex balance	High Q inductors
Schering Bridge	Capacitance	Complex balance	Capacitance, dissipation factor
Wien Bridge	Frequency	$R_1/R_2 = C_2/C_1$	Oscillators, frequency measurement

9. SIGNAL CONDITIONING

9.1 Signal Conditioning Operations

- **Amplification:** Increases signal amplitude (Op-Amp based amplifiers)
- **Attenuation:** Reduces signal amplitude for processing range
- **Filtering:** Removes unwanted frequency components (Low-pass, High-pass, Band-pass, Notch)
- **Linearization:** Compensates for non-linear sensor response
- **Isolation:** Provides electrical isolation between input and output
- **Conversion:** ADC (Analog to Digital), DAC (Digital to Analog)

9.2 Operational Amplifier Configurations

Inverting Amplifier

```
Gain (Av) = -Rf / Rin | Output = - (Rf/Rin) × Vin
```

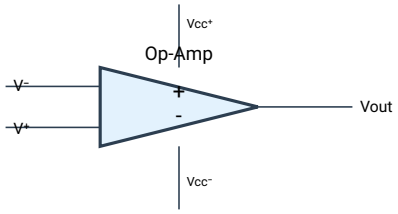
Non-Inverting Amplifier

```
Gain (Av) = 1 + Rf/Rin | Output = (1 + Rf/Rin) × Vin
```

Instrumentation Amplifier

- High input impedance, high CMRR (Common Mode Rejection Ratio)
- Used for amplifying small signals from sensors (strain gauges, thermocouples)

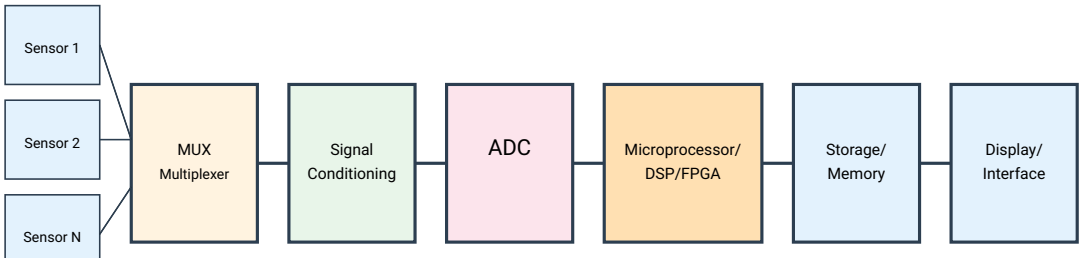
Figure 14: Operational Amplifier Symbol



10. DATA ACQUISITION SYSTEMS (DAQ)

A Data Acquisition System collects, processes, and stores data from multiple sensors.

Figure 15: Data Acquisition System Block Diagram

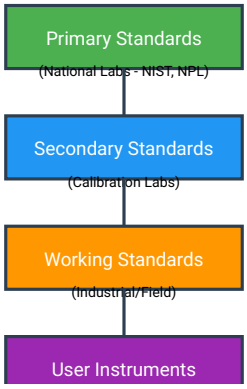


11. CALIBRATION & STANDARDS

11.1 Calibration Definition

Calibration is the process of comparing an instrument's output against a known standard and adjusting it to ensure accuracy.

11.2 Calibration Hierarchy



11.3 Calibration Terms

- **Traceability:** Unbroken chain of comparisons to national standards
- **Uncertainty:** Range within which true value lies (confidence interval)
- **Accuracy Class:** Classification based on percentage error
- **Calibration Interval:** Time between calibrations (typically 1 year)

12. MEASUREMENT OF PHYSICAL QUANTITIES

12.1 Temperature Measurement

Device	Range	Accuracy	Advantages	Disadvantages
Thermocouple	-200°C to 1800°C	±0.5°C to ±5°C	Wide range, rugged, fast response	Non-linear, reference junction needed
RTD (Pt100)	-200°C to 850°C	±0.1°C to ±0.5°C	High accuracy, stable, linear	Slow response, expensive
Thermistor	-50°C to 300°C	±0.1°C to ±1°C	High sensitivity, small size	Non-linear, limited range
Infrared Pyrometer	0°C to 3000°C	±1% of reading	Non-contact, fast	Emissivity error, expensive

12.2 Pressure Measurement

Device	Range	Principle	Application
Manometer (U-tube)	0-100 kPa	Hydrostatic balance	Low pressure, calibration
Bourdon Tube	0-1000 MPa	Elastic deformation	Industrial pressure gauges
Diaphragm	0-10 MPa	Deflection of diaphragm	Low to medium pressure

Piezoelectric	0-1000 MPa	Piezoelectric effect	Dynamic pressure, transients
Strain Gauge Pressure Sensor	0-100 MPa	Strain from pressure	General purpose, transmitters

12.3 Flow Measurement

Device	Principle	Application	Accuracy
Orifice Plate	Pressure drop across restriction	Liquids, gases, steam	±1% to ±3%
Venturi Meter	Pressure drop in converging section	Clean fluids, low pressure loss	±0.5% to ±1%
Rotameter	Float position vs flow	Small flow rates, visual	±2% to ±5%
Electromagnetic Flowmeter	Faraday's law (conductive fluids)	Conductive liquids	±0.5%
Ultrasonic Flowmeter	Doppler or transit time	Non-invasive, various fluids	±1% to ±5%

13. FORMULA SUMMARY

General:
Absolute Error = |Measured Value - True Value|
Relative Error = (Absolute Error / True Value) × 100%
Percentage Error = (Error / True Value) × 100%
Correction = True Value - Measured Value

Sensitivity: S = ΔOutput / ΔInput
Resolution: Least Count = (Range) / (Number of divisions)

Vernier Caliper:
Least Count = 1 MSD - 1 VSD = (Main Scale Division) / (Number of Vernier Divisions)
Reading = MSR + (VSD × LC)

Micrometer:
Least Count = Pitch / Number of divisions on thimble
Reading = Main Scale Reading + (Thimble Division × LC) ± Zero Error

Wheatstone Bridge:
At Balance: $R_1/R_2 = R_3/R_x \rightarrow R_x = (R_2 \times R_3)/R_1$

Strain Gauge:
$\Delta R/R = G \times \varepsilon$ where G = Gauge Factor, ε = Strain = $\Delta L/L$

Thermocouple:
$V = \alpha(T_1 - T_2)$ where α = Seebeck coefficient

Op-Amp Gains:
Inverting: Av = -Rf/Rin

```
Non-Inverting: Av = 1 + Rf/Rin

ADC Resolution:
Resolution = Vref / 2^n where n = number of bits

First Order System Response:
Output(t) = Input × (1 - e^(-t/τ))
Time Constant τ = RC or τ = L/R

Second Order System:
Natural Frequency ωn = √(k/m)
Damping Ratio ζ = c/(2√(km))
Overshoot Mp = e^(-πζ/√(1-ζ²)) × 100%
```

14. COMPARISON TABLES

14.1 Analog vs Digital Instruments

Parameter	Analog Instruments	Digital Instruments
Display	Pointer and scale	Numeric/LCD/LED
Accuracy	±0.5% to ±2%	±0.01% to ±0.5%
Reading Error	Parallax error possible	No parallax error
Resolution	Limited by scale divisions	High (up to 6½ digits)
Input Impedance	Low (loading effect)	High (minimal loading)
Cost	Lower	Higher
Response to transients	Good	Requires sampling rate consideration

14.2 Active vs Passive Transducers

Parameter	Active Transducers	Passive Transducers
Definition	Generates output without external excitation	Requires external excitation
Output	Voltage, charge	Resistance, capacitance, inductance
Examples	Thermocouple, piezoelectric, photovoltaic	Strain gauge, RTD, LVDT, thermistor
Signal Conditioning	Amplification only	Bridge circuit, excitation needed

14.3 Thermocouple Types

Type	Materials	Range (°C)	Application

Type K	Chromel-Alumel	-200 to 1260	General purpose, most common
Type J	Iron-Constantan	-40 to 750	Reducing atmosphere
Type T	Copper-Constantan	-200 to 350	Low temperature, cryogenic
Type E	Chromel-Constantan	-200 to 900	High output, cryogenic
Type S	Platinum-Rhodium	0 to 1480	High temperature, laboratory
Type R	Platinum-Rhodium	0 to 1480	High temperature, similar to S
Type B	Platinum-Rhodium	0 to 1700	Very high temperature

SUMMARY

Measurements and Instruments is a fundamental topic covering the principles, characteristics, and applications of various instruments used to measure physical quantities. Key concepts include:

- **Measurement systems** consist of sensor, signal conditioning, and display stages
- **Static characteristics** (accuracy, precision, sensitivity) define instrument performance
- **Dynamic characteristics** describe response to changing inputs
- **Errors** must be understood and minimized through calibration
- **Instruments** range from simple mechanical to advanced electronic systems
- **Transducers** convert physical quantities to electrical signals
- **Bridges** provide precise measurement of passive components
- **Calibration** ensures traceability and accuracy

END OF NOTES

Measurements and Instruments - Complete Reference